

Hindcasting the food-web dynamics of the tropical Indian Ocean pelagic ecosystem over the last two decades

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SUMMARY

Tuna species are among the most widely harvested fish stocks globally, contributing approximately 6.8% of the global finfish catches. In the tropical Indian Ocean, tropical tuna fisheries alone account for about 22% of the global tuna catches, however these fisheries also interact with a variety of non-targeted species, including vulnerable taxa. In the ecosystem, fisheries have the same ecological role as a high-trophic level predator, therefore it is important to monitor how their activity may influence the food web-structure and functioning of the oceanic pelagic ecosystem over time. In this study, we build upon the Ecopath model presented at the previous meeting of the IOTC Working Party on Ecosystems and Bycatch (Amate et al., 2024), updating biomass and discards estimates, refining dietary information, and incorporating expert knowledge review. Using this updated Ecopath model, we also developed and fitted a temporally dynamic Ecosim model to time series data from 2003 to 2022. The model reproduced satisfactorily observed trends for several functional groups, particularly those with biomass time series data available. For some vulnerable species and mid- to lower-trophic level groups, data limitations constrain the model's capacity to fully capture the historical ecosystem dynamics. Trophic interactions and fishing pressure emerge as key drivers of ecosystem change. Notably, results indicate a decline in the mean trophic level of the catch and in the biomass of commercially important species in line with those of the stock assessment. This study contributes to a deeper understanding of the dynamics of the tropical Indian Ocean pelagic ecosystem and marks a step forward in advancing the operationalization of the ecosystem approach to fisheries management in the IOTC.

KEYWORDS

EAFM, Ecopath, Ecosim, food web model, tuna, ecosystem, bycatch, ecological indicators, Indian Ocean

1. Introduction

Effective fisheries management is essential to ensure the long-term contribution of fisheries to global food security, livelihoods and the cultural identity of coastal communities and fishing nations closely tied to tuna resources (Boyd et al., 2022, Robinson et al., 2022). Among these fisheries, tunas and tuna-like species are among the most extensively harvested fish stocks globally, with catches reaching 5.2 million tonnes in 2023 (ISSF, 2025). Among the seven principal market tuna species, 87 percent of their total catches come from healthy stocks in terms of abundance (ISSF, 2025). However, tuna and tuna-like fisheries are not limited to catch the target species, they also interact with a variety of non-target species that may be incidentally caught as bycatch, including vulnerable taxa such as sharks and sea turtles. The

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combined removal of targeted and bycatch species may also influence the structure and functioning of pelagic food webs (Acevedo-Iglesias et al., 2025). Furthermore, climate change is reshaping marine ecosystems and fisheries through shifts in primary productivity, increasing acidification, and rising temperatures (Doney et al., 2012). These changes are already affecting marine species' distributions, phenology, and abundance, with direct impacts on fishing opportunities and dependent communities (Erauskin-Extramiana et al., 2023).

The Indian Ocean is a major fishing ground for tuna fisheries, but the Indian Ocean Tuna Commission (IOTC) faces significant challenges. The rapid expansion of both artisanal, semi-industrial and industrial fisheries led to catches peaking at 1.7 million tonnes in 2006. Since then, catches have slightly declined and have remained below that maximum (Murua et al., 2015). In the tropical Indian Ocean, bigeye tuna (*Thunnus obesus*), yellowfin tuna (*Thunnus albacares*), skipjack tuna (*Katsuwonus pelamis*) and small tunas (*Euthynnus affinis*, *Auxis thazard*, *Auxis rochei*) dominate the catch, of tuna and tuna-like fisheries. However, the catches in tuna fisheries are rarely monospecific, and bycatch of vulnerable species such as sharks and sea turtles is common (Huang and Liu, 2010, Ruiz et al., 2018, Elliott et al., 2024) with the bycatch species composition varying by fishing gear. Many of these bycatch species such as billfishes and sharks, together with the tunas, are top predators, playing key roles in regulating the structure and functioning of oceanic ecosystems (Kiszka et al., 2022, Dedman et al., 2024). Ecosystem variability due to the emerging pressure of climate change, incomplete catch and effort data reporting for some fisheries, poor decision-making processes and weak compliance monitoring are undermining the ability of the IOTC to ensure the long-term sustainability of tuna and tuna-like stocks, as well as the ecological integrity of the Indian Ocean pelagic ecosystem and the communities that depend on it (Erauskin-Extramiana et al., 2019, Welch et al., 2023).

IOTC have committed to operationalize the ecosystem approach to fisheries management (EAFM) (Andonegi et al., 2019, Juan Jordá, 2019). This holistic approach to fisheries management aims to complement traditional single-species management advice by incorporating bycatch, ecosystem and climate considerations. Within this context, marine ecosystem models are powerful tools as they provide a framework to integrate available information on ecosystem components, processes and anthropogenic pressures (fishing and climate change), providing a framework to explore system dynamics and inform fisheries management decisions (Geary et al., 2020). The Ecopath with Ecosim (EwE) modelling framework has been widely applied in both coastal and oceanic ecosystems (Colléter et al., 2015) including large-scale tuna fisheries (Olson and Watters, 2003, Forrestal and Ménard, 2016, Griffiths et al., 2018). EwE models are valuable tools for assessing the impact of fishing on marine ecosystems, evaluating the sustainability of resource exploitation, and exploring how environmental disturbances—such as climate change or species invasions—affect ecosystem dynamics. They also help researchers understand the structure and functioning of food webs, identifying key species and energy flows that maintain ecological balance (Corrales et al., 2017, Griffiths et al., 2018, Bornatowski et al., 2018).

This study presents the first Ecopath with Ecosim model developed specifically for the tropical Indian Ocean pelagic ecosystem to support decision-making processes within the Indian Ocean Tuna Commission (IOTC). Building on the static Ecopath model previously presented at the IOTC Working Party on Ecosystems and Bycatch (Amate et al., 2024), this study incorporates improved biomass and discards estimates, updated dietary information, and expert knowledge review. The temporally dynamic model Ecosim was then fitted to time series of data from 2003 to 2022, enabling simulation of ecosystem responses over time. Using this EwE model, the objective of this study is to improve the understanding of the structure and the functioning of the Tropical Indian Ocean (TIO) oceanic pelagic ecosystem, by characterizing its key trophic interactions and its historical dynamics in response to fishing between 2003 and 2022. Subsequently, this model will be used to assess the cumulative impacts of fishing on tuna and tuna-like species, as well as the broader ecosystem, under both historic and potential future fishing and climate change scenarios. Model outputs will include ecological indicators that connect the impacts of fishing and climate change on species and ecosystem structure and functioning to fisheries management advice. These outputs will directly support the current development of EAFM products (e.g. regional EcoCard and Ecosystem Fisheries Overviews) within IOTC, providing managers with a broader ecological context for managing tropical tuna species and associated ecosystems in the tropical Indian Ocean.

2. Methods

2.1. Ecopath model

The Tropical Indian Ocean (TIO) Ecopath model is a quantitative representation of trophic flows and biomasses in the pelagic ecosystem. The model represents the mean state of the ecosystem between 2000 and 2003 and covers an area of over 21.7 million km² between 0 and 500 m depth (Figure 1). The model represents oceanic waters, excluding the continental shelf from the analysis (except for small islands and archipelagos that extend within the study area), since it represents an area of minor importance for tropical tuna fisheries and because including it would imply modelling the demersal/benthic ecosystem, as well as the inclusion of other more artisanal and national fleets. In both cases, this would have an irrelevant impact on the dynamics of the target species in ecological and fishery terms, while simultaneously adding greater uncertainty to the model.

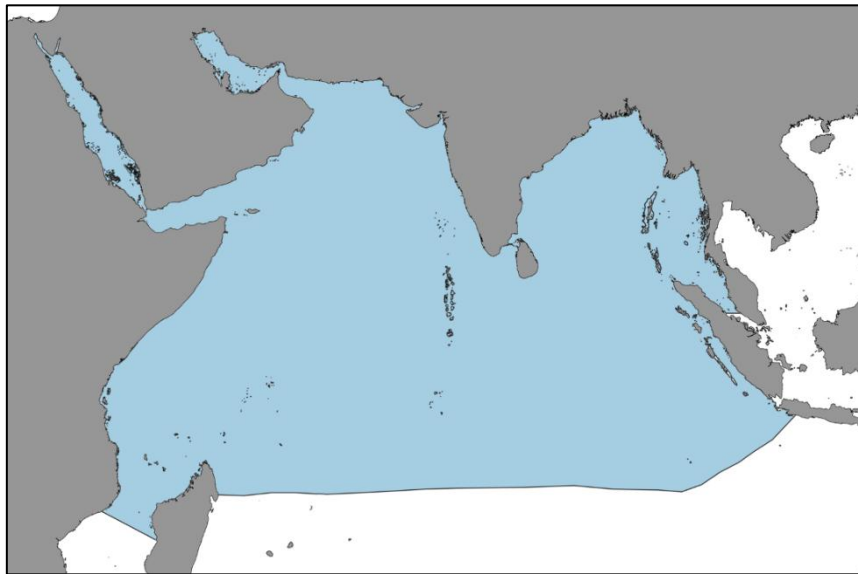


Figure 1. Practical model area (in blue). The theoretical area represents the same area of the tropical Indian Ocean, but without including the continental shelf (except for small islands and archipelagos).

The model is composed of 35 functional groups (FG), ranging from primary producers to top predators. Functional groups represent ontogenetic stages of a species (e.g., mature and immature), single species, or groups of species that share ecological characteristics (e.g., diet) (Christensen and Walters, 2004). Functional groups of the TIO model were defined according to the commercial importance of the species, data availability, and the ecology of the species, such as diet or bathymetric distribution (Table 1). The model includes ten groups of vulnerable species (one for seabirds, two for cetaceans, six for elasmobranchs, and one for sea turtles). Main target species were modelled separately such as swordfish (*Xiphias gladius*) or albacore tuna (*Thunnus alalunga*). In addition, the three tropical tuna species: bigeye tuna (BET), yellowfin tuna (YFT), and skipjack tuna (SKJ) were separated into two ontogenetic groups (i.e., matures and immatures) attaining their length at first maturity for fishery and ecological reasons. The model also includes functional groups representing species with the highest incidental catch rates by tuna fleets (e.g., silky shark, other sharks, mobulas).

Estimates of discards were obtained from the literature, except for the Spanish, French and Seychellois purse seine fleets. For these fleets, onboard observers' data from AZTI and IRD was used to estimate the discards in weight (t) of a species or functional group per 1000 tonnes of targeted tuna landed (BET, YFT, SKJ). Regarding discard survival, two distinct periods were assumed for these fleets: a period prior to 2014 when handling and release best practices were not yet implemented onboard tuna vessels and from 2014 onwards, when such best practices were being implemented (Jon Ruiz, *pers. comm.*). For the pre-2014 period, discard mortality was assumed to be 100%. In contrast, for the period post-2014, discard mortality was estimated based on empirical studies of at-haulback mortality and post-release survival

(Table 3). Then, for the remaining fleets, discard survival rates were calculated using published post-release and at-haulback mortality rates, which were applied consistently across the entire simulated period. (Tables 2 and 3).

Table 1. Input data and outputs of the Ecopath TIO model. Parameters estimated by the model are highlighted in bold. B = final biomass (t-km⁻²); P/B = production/biomass (year⁻¹); Q/B = consumption/ biomass (year⁻¹); P/Q = production/consumption ratio; EE = ecotrophic efficiency; TL = trophic level; F = fishing mortality; F/Z=exploitation rate.

Functional group	B	P/B	Q/B	P/Q	EE	TL	F	F/Z
1 Seabirds	0.0005	0.12	77.62	0.00	0.01	4.19	-	-
2 Baleen whales	0.0152	0.04	3.98	0.01	0.48	3.86	0.017	0.447
3 Toothed whales	0.0460	0.06	8.16	0.01	0.43	4.28	0.004	0.077
4 Blue shark	0.0145	0.40	2.92	0.14	0.66	4.54	0.190	0.471
5 Silky shark	0.0200	0.44	4.45	0.10	0.30	4.09	0.115	0.259
6 Other sharks	0.0240	0.34	3.06	0.11	0.67	4.57	0.211	0.624
7 Whale shark	0.0430	0.04	1.57	0.03	0.50	3.42	0.020	0.469
8 Pelagic stingray	0.0176	0.59	6.57	0.09	0.30	3.80	0.002	0.004
9 Mobulas	0.0012	0.38	3.79	0.10	0.57	3.82	0.149	0.398
10 Sea turtles	0.0017	0.19	3.04	0.06	0.64	3.73	0.094	0.508
11 Other billfishes	0.0143	0.33	3.48	0.01	0.41	4.46	0.100	0.301
12 Swordfish	0.0103	0.31	2.57	0.12	0.35	4.30	0.098	0.317
13 Bigeye mature	0.0262	0.32	2.85	0.11	0.71	4.51	0.214	0.669
14 Bigeye immature	0.0127	0.52	4.84	0.11	0.74	4.45	0.142	0.273
15 Yellowfin mature	0.0795	0.41	3.68	0.11	0.41	4.26	0.157	0.385
16 Yellowfin immature	0.0306	0.60	5.45	0.11	0.66	3.97	0.121	0.202
17 Skipjack mature	0.0671	0.51	3.97	0.13	0.80	3.90	0.287	0.565
18 Skipjack immature	0.0023	1.05	8.76	0.12	0.89	3.86	0.268	0.254
19 Albacore	0.0027	0.66	4.25	0.16	0.51	4.15	0.096	0.144
20 Small tunas and bonitos	0.1836	0.79	6.12	0.13	0.70	3.75	0.085	0.107
21 Jacks, mackerels, runners, trevallies and scads	0.1609	1.01	7.76	0.13	0.80	3.61	0.002	0.002
22 Epipelagic fishes - I	0.1532	0.84	6.44	0.13	0.70	3.85	0.002	0.002
23 Epipelagic fishes - II	0.1929	0.95	6.79	0.14	0.80	3.57	0.000	0.000
24 Small epipelagic fishes	0.5532	2.09	12.24	0.17	0.73	3.26	0.000	0.000
25 Large mesopelagic fishes	0.2225	0.51	3.93	0.13	0.80	4.12	0.000	0.000
26 Small mesopelagic fishes	2.6816	2.71	13.96	0.19	0.68	2.75	-	-
26 Pelagic cephalopods	0.4732	2.43	15.87	0.15	0.90	3.57	-	-
27 Pelagic crustaceans	0.8220	5.00	17.00	0.29	0.62	2.54	-	-
29 Gelatinous plankton	0.5395	4.75	23.80	0.20	0.85	2.95	-	-
30 Macrozooplankton	2.6720	17.75	62.86	0.28	0.24	2.73	-	-
31 Mesozooplankton	4.7460	33.79	121.7	0.28	0.58	2.35	-	-
32 Microzooplankton	7.0020	61.21	204.6	0.30	0.62	2.02	-	-
33 Phytoplankton	11.658	204.8	-	-	0.60	1.00	-	-
34 Discards	0.0110	-	-	-	0.22	1.00	-	-
35 Detritus	238.88	-	-	-	0.27	1.00	-	-

Furthermore, the model includes 13 fleets defined by fishing gear, i.e., differentiating between purse seine, longline, baitboat, gillnet, line, and other gears. Purse seine fishing was divided into two types: purse seine fleets setting on free-schools (FSC) and purse seine fleets setting on objects (FAD). This distinction was made to capture differences in catch composition, although the same vessel can make both types of sets. Furthermore, purse seine fleets were divided into four groups: Spain, France, Seychelles, and Others due to the availability of data for these fleets. Others – purse seine fleets include all other purse seine fleets operating in the study area excluding the mentioned flags. For a more detailed description of the Ecopath TIO model, see Amate et al. (2024).

Table 2. References used to calculate the discard rate by fleet and functional group.

Functional group/Fleet	Gillnet	Longline	Purse seine – FSC & FAD (ESP, FRA, SYC)	Purse seine – FSC & FAD (Others)
2. Baleen whales	(Garcia and Herrera, 2018)	(Garcia and Herrera, 2018)		
3. Toothed whales	(Anderson et al., 2020)	(Garcia and Herrera, 2018)		
4. Blue Shark	(Garcia and Herrera, 2018)	(Garcia and Herrera, 2018)	Observers' data	(Ruiz et al., 2018)
5. Silky shark	(Garcia and Herrera, 2018)	(Garcia and Herrera, 2018)	Observers' data	(Ruiz et al., 2018)
6. Other sharks	(Garcia and Herrera, 2018)	(Garcia and Herrera, 2018)	Observers' data	(Ruiz et al., 2018)
7. Whale shark	(Garcia and Herrera, 2018)			(Garcia and Herrera, 2018)
8. Pelagic stingray	(Garcia and Herrera, 2018)	(Garcia and Herrera, 2018)		
9. Mobulas	(Garcia and Herrera, 2018)	(Garcia and Herrera, 2018)		
10. Sea turtles	(Garcia and Herrera, 2018)	(Garcia and Herrera, 2018)		
11. Other billfishes		(Huang and Liu, 2010)	Observers' data	(Ruiz et al., 2018)
12. Swordfish		(Huang and Liu, 2010)	Observers' data	(Ruiz et al., 2018)
13. Bigeye mature		(Huang and Liu, 2010)		
14. Bigeye immature		(Huang and Liu, 2010)	Observers' data	(Ruiz et al., 2018)
15. Yellowfin mature				
16. Yellowfin immature		(Huang and Liu, 2010)	Observers' data	(Ruiz et al., 2018)
17. Skipjack mature				
18. Skipjack immature			Observers' data	(Ruiz et al., 2018)
19. Albacore		(Huang and Liu, 2010)	Observers' data	(Ruiz et al., 2018)
20. Small tunas and other bonitos			Observers' data	(Ruiz et al., 2018)
21. Jacks, mackerels, runners, trevallies and scads	(Shahifar, 2018)		Observers' data	(Ruiz et al., 2018)
22. Epipelagic fishes – I	(Shahifar, 2018)		Observers' data	(Ruiz et al., 2018)
23. Epipelagic fishes - II			Observers' data	(Ruiz et al., 2018)
24. Small epipelagic fishes			Observers' data	(Ruiz et al., 2018)
25. Large mesopelagic fishes			Observers' data	

Table 3. References used to calculate the discard mortality rate (DMR) by functional group and fleet based on estimates of post-release mortality rate (PRM), at-haulback mortality rate (AHM), and at-haulback survival rate (AHS).

Functional group	Fleet	DMR	PRMr	PRM – Reference	AHM	AHM – Reference	AHS
Seabirds	Baitboat	0	0	(Miller et al., 2017)	0	(Miller et al., 2017)	1
Seabirds	Gillnet	0	0	-	0	(Moazzam and Nawaz, 2014)	1
Baleen whales	Purse seine (ESP, FRA, SYC)	0	0	-	0	(Ruiz et al., 2018)	1
Baleen whales	Gillnet	1	0	-	1	(Moazzam and Nawaz, 2014)	0
Baleen whales	Longline	0	0	-	0	(Sabarros et al., 2013)	1
Toothed whales	Purse seine (ESP, FRA, SYC)	0	0	-	0	(Escalle et al., 2015)	1
Toothed whales	Gillnet	1	0	-	1	(Moazzam and Nawaz, 2014)	0
Toothed whales	Longline	0.09	0	-	0.09	(Sabarros et al., 2013)	0.91
Blue shark	Purse seine (ESP, FRA, SYC)	0.57	0	-	0.57	(Clavareau et al., 2020)	0.43
Blue shark	Longline	0.42	0.23	(Campana et al., 2009)	0.25	(Coelho et al., 2011)	0.75
Silky shark	Purse seine (ESP, FRA, SYC)	0.62	0.41	(Grande et al., 2022)	0.35	(Clavareau et al., 2020)	0.65
Silky shark	Baitboat	0.18	0	-	0.18	(Miller et al., 2017)	0.82
Silky shark	Longline	0.79	0.2	(Musyl and Gilman, 2018)	0.74	(Coelho et al., 2011)	0.26
Other sharks	Purse seine (ESP, FRA, SYC)	0.89	0.85	(Eddy et al., 2016)	0.28	(Clavareau et al., 2020)	0.72
Other sharks	Longline	0.64	0.14	(Hutchinson et al., 2021)	0.58	(Coelho et al., 2011)	0.42
Whale shark	Gillnet	0.20	0	-	0.20	(Moazzam and Nawaz, 2014)	0.80
Whale shark	Purse seine (ESP, FRA, SYC)	0	0	(Grande et al., 2019)	0	(Capietto et al., 2014)	1
Pelagic stingray	Purse seine (ESP, FRA, SYC)	0.64	0	-	0.64	(Clavareau et al., 2020)	0.36
Pelagic stingray	Longline	0	0	-	0	(Coelho et al., 2011)	1
Mobulas	Purse seine (ESP, FRA, SYC)	0.91	0.83	(Clavareau et al., 2020)	0.44	(Stewart et al., 2024)	0.56
Mobulas	Longline	0	0	-	0	(Coelho et al., 2011)	1
Sea turtles	Longline	0.69	0.28	(Swimmer et al., 2013)	0.57	(Huang and Liu, 2010)	0.43
Sea turtles	Purse seine (ESP, FRA, SYC)	0.25	0.13	(Bourjea et al., 2014)	0.14	(Bourjea et al., 2014)	0.86
Sea turtles	Gillnet	0.05	0	-	0.05	(Moazzam and Nawaz, 2014)	0.95
Other billfishes	Purse seine (ESP, FRA, SYC)	0.93	0	-	0.93	(Amandè et al., 2008)	0.07
Other billfishes	Longline	0.83	0.64	(West et al., 2012)	0.52	(Li et al., 2023)	0.48
Swordfish	Longline	0.87	0.64	(West et al., 2012)	0.64	(Guo et al., 2022)	0.36

2.2. Ecosim model

The static Ecopath model was calibrated and fitted to available time series data from 2003 to 2022 using Ecosim. Ecosim is the time-dynamic module of the EwE approach, built on a set of differential equations that describe biomass dynamics over time (Christensen and Walters, 2004, Walters et al., 1997). These equations are derived from the fundamental Ecopath production equation and are expressed as follow:

$$\frac{dB_i}{dt} = g_i \sum Q_{ji} - \sum Q_{ij} + I_i - (M0_i + F_i + e_i) \times B_i$$

Where $\frac{dB_i}{dt}$ represents the rate of biomass change, B_i , of the functional group i during the time interval dt . g_i is the net growth efficiency, Q_{ji} is the consumption rate of j by the functional group i , and Q_{ij} is the consumption rate on the functional group i by the predator j . I_i is the immigration rate, while $M0_i$, F_i , e_i represent the natural mortality rate, fishing mortality rate, and emigration rate, respectively. When multiplied by B_i , these yield the net rates of the three factors.

Consumption rates (Q) are estimated using the “foraging arena theory” (Ahrens et al., 2011). This theory is based on the idea that organisms are not constantly exposed to predation risks, as they employ various strategies to reduce it, such as hiding, schooling, or performing vertical migrations. The “foraging arena theory” defines two states: *vulnerable*, when the prey is available to predators, and *non-vulnerable*, when the prey is not accessible to predators. In the model this implies the division of the prey’s biomass between a vulnerable and a non-vulnerable fraction, and the transfer rate or vulnerability (v_{ij}) between the two fractions determines the trophic flow between the predator and the prey (Ahrens et al., 2011). This concept of vulnerability incorporates density-dependent processes and reflects how close a species or functional group is to its carrying capacity. Carrying capacity refers to the maximum population size that a species or group can reach without negatively impacting itself due to the depletion of essential resources required for its own development (Christensen and Walters, 2004, Christensen et al., 2008). The default vulnerability values ($v_{ij} = 2$) represents a mixed trophic flow. A low value ($v_{ij} < 2$) indicates a “bottom-up” flow and a situation closer to the carrying capacity, whereas a high value ($v_{ij} > 2$) suggests a “top-down” flow and a condition further from the carrying capacity (Ahrens et al., 2011). For each predator-prey interaction, the consumption rate is calculated as follows:

$$Q_{ij} = \frac{a_{ij} \times v_{ij} \times B_i \times B_j \times T_i \times T_j \times M_{ij}/D_j}{v_{ij} + v_{ij} \times T_i \times M_{ij} + a_{ij} \times M_{ij} \times B_j \times T_j/D_j} \times f(Env_{function}, t)$$

Where a_{ij} is the effective search rate of predator j on prey i , v_{ij} is the vulnerability of prey i by predation of j , B_i is the biomass of the prey, B_j is the biomass of the predator, T_i is the relative feeding time of the prey, and T_j is the relative feeding time of the predator. M_{ij} represents mediation effects, and D_j denotes the handling time of the prey by the predator, which limits the consumption rate. $f(Env_{function}, t)$ is a multiplier factor that accounts for the influence of environmental variables (e.g., temperature). This multiplier ranges from 0 to 1 and is derived from the environmental response function, modifying the consumption rate (Q_{ij}) of the functional group. Specifically, the multiplier is calculated based on the intersection between the response function to an environmental factor and the actual conditions of that factor, with a maximum value of 1 and decreasing values as the environmental condition deviates from the optimal range for the functional group.

2.3. Time series

Time series of fishing effort, fishing mortality, and primary production (PP) were used to drive the model, while time series of biomass and catches were used to calibrate the model, comparing model’s predictions with observed data (Table 4).

Biomass time series for targeted species were obtained from fishery stock assessments conducted by IOTC. Additionally, outputs from global ocean models were used to derive biomass time series for lower trophic level functional groups (Table 4). Catch time series were calculated based on the official IOTC landings database. See Amate et al. (2024) for a more detailed description of the calculations.

Time series of fishing effort were calculated using the official IOTC catch and effort database or obtained from published sources (Table 4, Table 5, Figure 2 – Supplementary material). For the free-school (FSC) and setting on objects (FAD) purse seine fleets of Spain, France, and Seychelles, effort time series in sets were extracted from published studies and for some years, by personal communication (Table 4). For the “Others – FSC purse seine” and ‘Others – FAD purse seine’ fleets, where no fishing effort data were available, effort time series in sets were estimated by weighting the effort of the other three purse seine fleets in the model (Spain, France, and Seychelles) according to their total catches to obtain two CPUE time series. The effort time series were obtained by dividing the catches of each fleet by their corresponding CPUE. For the rest of the fleets (defined by flag and gear type) included in the model—namely baitboat, others, line, gillnet, and longline—an analysis was conducted to assess the quality and temporal coverage of effort data reported in the IOTC database. Fleets with the highest catch volumes within each fleet grouping included in the model and with the most reliable effort data were selected (Table 5). For the baitboat fleet, 90% of the catches correspond to the Maldivian fleet (Table 5). The effort trend of the baitboat fleet was assumed to follow the same pattern as that of the Maldivian baitboat fleet, scaled according to total catches. Since the effort of the Maldivian baitboat fleet was reported in trips and fishing days in the IOTC catch and effort database, trips were converted into fishing days using a conversion factor of 1 trip = 1 fishing day (*pers. comm.*). Regarding the gillnet fleet, the two main contributors are the Iranian and Sri Lankan fleets, which together account for nearly 80% of the gillnet fleet catches in the model (Table 5). As both fleets report their effort in trips, a CPUE for each fleet was calculated and then combined into a weighted CPUE based on their respective catches. Using the total catches of the gillnet fleet and the combined CPUE, the data to estimate the effort time series were scaled. For the longline fleet, the effort data from the Chinese, Taiwanese, and Japanese fleets, which together represent over 55% of the longline fleet catches in the model were combined. All three fleets report their effort in hooks (Table 5). By combining their effort and catch data, a CPUE was derived, which was then used to scale the total catches of the longline fleet to approximate its overall effort. For the line and other fleets, no effort data were available to calculate their effort time series. Therefore, a relative change in effort based on the variation in catches was assumed using the year 2003 as the reference point (Table 4).

Time series of exploitation rates (catch/biomass) were incorporated for species with available fishery stock assessments to represent fishing impacts (Figure 2 – supplementary material).

In addition, the model includes a time series of primary production (PP) (Table 2; Figure 3 – supplementary material), obtained from the Copernicus Global Ocean Biogeochemistry Hindcast product (<https://doi.org/10.48670/moi-00019>), as changes in primary production can have a significant impact on the ecosystem.

To fit the Ecosim model to the observed time series data from 2003 to 2022, we used the stepwise fitting procedure described by Scott et al. (2016). In our model, this procedure estimates various vulnerability parameters to improve model calibration by comparing model predictions with observed data, using the sum of squares (SS) statistic. The fitting procedure identifies the model that provides the best statistical fit based on the Akaike Information Criterion (AIC), which penalizes model over-parameterization (Heymans et al., 2016, Mackinson et al., 2009). The maximum number of vulnerabilities that can be estimated is $k-1$, where k is the number of observed time series (in this case 40 time series of biomass and catch). Finally, to choose the best final model, we manually evaluated whether the parameterization process produced credible and sensible behavior (Heymans et al., 2016). From the three models with the best overall fit, the final model was chosen based on how well it reproduced observed values of the three tropical tuna species.

Once the calibration was completed, changes in the biomass and catch of selected functional groups were analysed, considering both their ecological and economic importance. To assess the correlation between model outputs (biomass and catches) and time, the Spearman’s rank correlation coefficient was used.

Table 4. Time series used to drive, adjust and calibrate the TIO Ecosim model. ESP: Spain; FRA: France; SYC: Seychelles; BUM: Blue marlin; BLM: Black marlin; MLS: striped marlin; SFA: Sailfish

	Data time series	Coverage	References	Drive the model	Adjust the model	Compare predicted results
4. Blue shark	Absolute biomass	2003-2019	Stock assessment (IOTC, 2022c)		X	X
	Absolute catch	2003-2022	IOTC - Nominal Catch (accessed. 22-02-2024)		X	X
	Harvest rate (F)	2003-2019	Calculated	X		
2. Baleen whales; 3. Toothed whales; 5. Silky shark 6. Other sharks; 7. Whale shark; 8. Pelagic stingray 9. Mobulas; 10. Sea turtles 20. Small tunas and other bonitos 21. Jacks, mackerels, runners, trevallies and scads 22. Epipelagic fishes - I; 23. Epipelagic fishes - II 25. Large mesopelagic fishes	Absolute catch	2003-2022	IOTC - Nominal Catch (accessed. 22-02-2024), estimates from the literature and onboard observers' data.		X	X
11. Other billfishes	Absolute biomass	2003- 2019, 2020 and 2022	BUM - (Parker and Kerwath, 2022); BLM - (Chih-Yu et al., 2024); MLS - (Yun-ju et al., 2024); SFA - (Parker and Espittia, 2022);		X	X
	Absolute catch	2003-2022	IOTC - Nominal Catch (accessed. 22-02-2024)		X	X
	Harvest rate (F)	2003- 2019, 2020 and 2022	Calculated	X		
12. Swordfish	Absolute biomass	2003-2022	Stock assessment (IOTC, 2022d)		X	X
	Absolute catch	2003-2022	IOTC - Raised catch (accessed. 13-10-2023)		X	X
	Harvest rate (F)	2003-2022	Calculated	X		
13. Bigeye mature 14. Bigeye immature	Absolute biomass	2003-2022	Stock assessment (IOTC, 2022b)		X	X
	Absolute catch	2003-2022	IOTC - Raised catch (accessed. 13-10-2023)		X	X
	Harvest rate (F)	2003-2022	Calculated	X		
15. Yellowfin mature 16. Yellowfin immature	Absolute biomass	2003-2022	Stock assessment (IOTC, 2021)		X	X
	Absolute catch	2003-2022	IOTC - Raised catch (accessed. 13-10-2023)		X	X
	Harvest rate (F)	2003-2022	Calculated	X		
17. Skipjack mature 18. Skipjack immature	Absolute biomass	2003-2022	Stock assessment (IOTC, 2023)		X	X
	Absolute catch	2003-2022	IOTC - Raised catch (accessed. 13-10-2023)		X	X
	Harvest rate (F)	2003-2022	Calculated	X		
19. Albacore	Absolute biomass	2003-2022	Stock assessment (IOTC, 2022a)		X	X
	Absolute catch	2003-2022	IOTC - Raised catch (accessed. 13-10-2023)		X	X
	Harvest rate (F)	2003-2022	Calculated	X		
24. Small epipelagic fishes 26. Small mesopelagic fishes	Absolute biomass	2003-2022	LMTL model (CMEMS - Copernicus) (Lehodey et al., 2015)		X	X
30. Macrozooplankton; 31. Mesozooplankton 32. Microzooplankton	Absolute biomass	2003-2022	COPEPOD database and (Wimalasiri et al., 2021)		X	X
33. Phytoplankton	Absolute biomass	2003-2022	Global Ocean Biogeochemistry Hindcast (CMEMS - Copernicus)		X	X
	Relative fishing effort (ESP – Purse seine fleets)	2003-2019	(Báez et al., 2020) & IOTC - Catch & Effort (accessed. 02-04-2025)	X		
	Relative fishing effort (FRA – Purse seine fleets)	2003-2022	(Floc'h et al., 2021) & (Laurent Floc'h, <i>pers. comm.</i>)	X		
	Relative fishing effort (SYC – Purse seine fleets)	2003-2022	(Assan et al.) & (Cindy Assan, <i>pers. comm.</i>)	X		
	Relative fishing effort (Others – Purse seine fleets)	2003-2022	Approximated using the effort of the other purse seine fleets	X		
	Relative fishing effort (Baitboat, Gillnet and Longline fleets)	2003-2022	IOTC - Catch & Effort (accessed. 14-10-2024)	X		
	Relative fishing effort (Line and Others fleets)	2003-2022	Follow the catch trends - IOTC - Raised catch (accessed. 13-10-2023)	X		
	Environmental variable – Primary production (PP)	2003-2022	Global Ocean Biogeochemistry Hindcast (CMEMS - Copernicus)	X		

Table 5. Information on the fleets used to calculate the effort time series for the baitboat, gillnet and longline fleets in the model. The % of catches refer to the proportion of total catches by that flag relative to the total catches of the fleet between 2000 and 2003. TS units refer to the units used in the time series of the effort.

Fleet	Flag	Catches %	Period	Effort units	TS units	Dataset
Baitboat	Maldives	90.1	2004-2012 2013-2022	TRIPS FDAYS	FDAYS	IOTC-2024-WPB22-DATA05-CESurface (https://iotc.org/data/datasets/latest/CE/Surface)
Gillnet	Iran	46.3	2007-2022			
	Sri Lanka	32.5	2000-2006 2014-2022	TRIPS	TRIPS	IOTC-2024-WPB22-DATA06-CEOther (https://iotc.org/data/datasets/latest/CE/All)
Longline	Taiwan	40.7	2000-2022			
	China	5.5	2000-2022	HOOKS	HOOKS	IOTC-2024-WPB22-DATA04-CELlongline (https://iotc.org/data/datasets/latest/CE/Longline)
	Japan	9.1	2000-2022			

Finally, to evaluate the uncertainty in Ecopath input parameters (biomass, production, consumption, ecotrophic efficiency, and catches) on Ecosim outputs (biomass, catches, and ecological indicators), the Monte Carlo (MC) simulation routine was applied. This routine modifies various Ecopath input parameters within the confidence intervals previously defined in the pedigree and checks whether the modified values meet the mass-balance criteria. If the criteria are met, the MC routine runs the Ecosim model using the adjusted Ecopath configuration and the parameters established during the original model calibration. The outputs from the various Ecosim simulations are then used to assess the range of possible trajectories in biomass and catches of the functional groups, thereby determining the level of uncertainty. Specifically, for this model, 250 Monte Carlo simulations were conducted based on the pedigree index previously introduced in the Ecopath model of the TIO to determine the 5th and 95th confidence intervals for the Ecosim results.

2.4. Ecological indicators

A selection of ecological indicators was used to describe ecological changes in the ecosystem over time. To calculate these indicators, the ECOIND and the Ecological Network Analysis (ENA) plug-ins of EwE were used (Coll and Steenbeek, 2017). The indicators selected for this analysis were:

- (1) **Predatory biomass ($t \cdot km^{-2}$):** Biomass of all functional groups with a trophic level (TL) ≥ 3.95 . This indicator typically declines with increasing fishing pressure (Rochet and Trenkel, 2003).
- (2) **Commercial species biomass ($t \cdot km^{-2}$):** Biomass of all functional groups subject to catches in the model. Biomass falling under certain levels may refer to overfished stocks.
- (3) **Mean trophic level of the community (mTLco):** Average TL of the entire ecosystem, widely used to assess fishing impacts (Rochet and Trenkel, 2003). The mTLco typically decreases as fishing pressure increases.
- (4) **Kempton's Q index:** A measure of biomass diversity in the ecosystem for organisms with TL ≥ 3 . High values may indicate a healthy and balanced ecosystem, while declines may signal ecosystem degradation (Kempton and Taylor, 1976).
- (5) **Mean trophic level of the catch (mTLc):** Average TL of the catches across all fleets combined. Used to evaluate fishing strategies and their impacts (Pauly et al., 1998). mTLc falling under certain levels may point to overexploitation of top predators
- (6) **Mean trophic level of the community > 2 (mTLco > 2):** Average TL of all the functional groups with a TL higher than 2 and lower than 3.25. Used to track changes in lower trophic levels of the ecosystem (Rochet and Trenkel, 2003).
- (7) **Mean trophic level of the community > 3.25 (mTLco > 3.25):** Average TL of all the functional groups with a TL higher than 3.25 and lower than 4. Used to track changes in intermediate trophic level species of the ecosystem (Rochet and Trenkel, 2003)

- (8) **Mean trophic level of the community > 4 (mTL_{co} > 4)**: Average TL of all the functional groups with a TL larger than 4. Used to track changes of top predators (Rochet and Trenkel, 2003).

The uncertainty associated with the input parameters of Ecopath, as previously described, also enables the assessment of uncertainty in ecological indicators (Coll and Steenbeek, 2017).

3. Preliminary results

3.1. Adjustment of the Ecosim model to the historical period (2003-2022)

The Ecosim model satisfactorily adjusted biomass predictions to the trends of observed values for most target species (Figure 2). The model has successfully reproduced the observed historical trends of the biomass of the blue shark (FG. 4), other billfishes (FG. 11), bigeye mature (FG. 13), bigeye immature (FG. 14), yellowfin mature (FG. 15), yellowfin immature (FG. 16), skipjack mature (FG. 17) and albacore (FG. 19) as well as small epipelagic fishes (FG. 24). However, the model failed to adequately fit the trends of observed biomass values for swordfish (FG. 12), skipjack immature (FG. 18) and small mesopelagic fishes (FG. 26) (Figure 2).

The model predicts declining biomass trends for several functional groups targeted by fisheries (Figure 2). For example, the biomasses of mature and immature bigeye and yellowfin tuna, other billfishes and albacore exhibit a negative trend during the simulated period. Additionally, other functional groups, such as mature and immature skipjack tuna and blue shark, exhibit a negative trend at the beginning of the simulated period, followed by a stable trend with a slight increase toward the end of the simulation. (Figure 2). Similarly, the model predicts significant reductions in most functional groups of vulnerable species, such as silky shark, other sharks, mobulas, sea turtles, baleen whales, toothed whales, whale shark, and pelagic stingray. However, the uncertainty surrounding the predictions for these vulnerable functional groups is considerably higher (Figure 4 – supplementary material). On the contrary, the model predicts stable or even increasing biomass trends for intermediate functional groups like, small tunas and other bonitos, jacks, mackerel, runners, trevallies, and scads, and epipelagic fishes – I, II (Figure 4 – supplementary material).

For the 19 functional groups without time series of biomass observations, the model has also predicted biomass trends throughout the entire historical period (Figure 4 – supplementary material).

The model accurately reproduces the biomass declines observed in the assessments of bigeye and yellowfin tuna, reflecting the exploitation experienced by both species during the simulated period. It also correctly reproduces the post-2008 stabilization of mature and immature skipjack tuna biomasses (Figure 2), indicating that their biomasses are approaching the targeted B_{MSY} levels. Additionally, the model shows a declining trend for silky shark, other sharks, and mobulas, as these are among the main non-target functional groups impacted by fishing in the area (Figure 4 – supplementary material).

Regarding catches, the model satisfactorily fitted most of the trends of observed values for most functional groups (Figure 3), particularly those with available time series of fishing mortality. Specifically, the model reproduced the declining trend in the catches of blue shark, mature and immature bigeye tuna, mature yellowfin tuna, and albacore. It also shows an increase in the catches of other billfishes, immature yellowfin tuna and mature and immature skipjack tuna (Figure 3).

Time series of observed catch values were available for 12 of the 19 groups without biomass observations. Among these, the model was able to fit the catch time series for the silky shark (FG. 5), epipelagic fishes – I (FG. 22) and small epipelagic fishes (FG. 24) (Figure 5 – supplementary material). By contrast, it couldn't fit the catch trends for the baleen whales (FG. 2), toothed whales (FG. 3), other sharks (FG. 6), whale shark (FG. 7), pelagic stingray (FG. 8), mobulas (FG. 9), sea turtles (FG. 10), small tunas and other bonitos (FG. 20), jacks, mackerels, runners, trevallies and scads (FG. 21), epipelagic fishes – II (FG. 23) and large mesopelagic fishes (FG. 25) (Figure 5 – supplementary material). Overall, catch trends of non-targeted species show an increase in the area due to the general rise in fishing effort throughout the entire period (Figure 1 & Figure 5 – supplementary material).

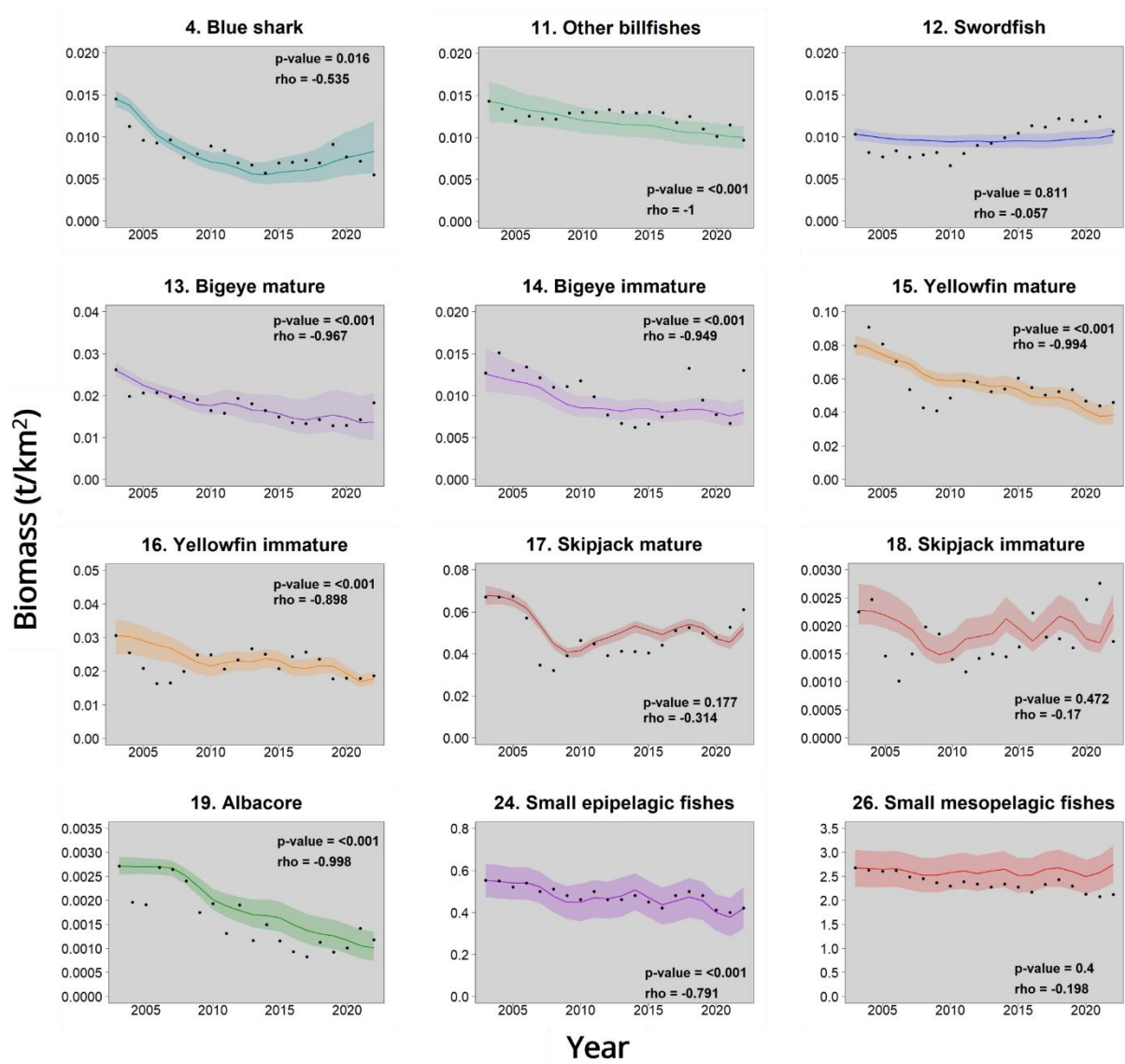


Figure 2. Predicted (solid line) versus observed (black dots) values of biomass (t/km²) of the functional groups with biomass time series for the period 2003 – 2022 in the TIO model. Coloured areas around the prediction line represent the 5th and 95th percentiles obtained by the Monte Carlo routine. The p-value and rho are the results of the Spearman correlation test.

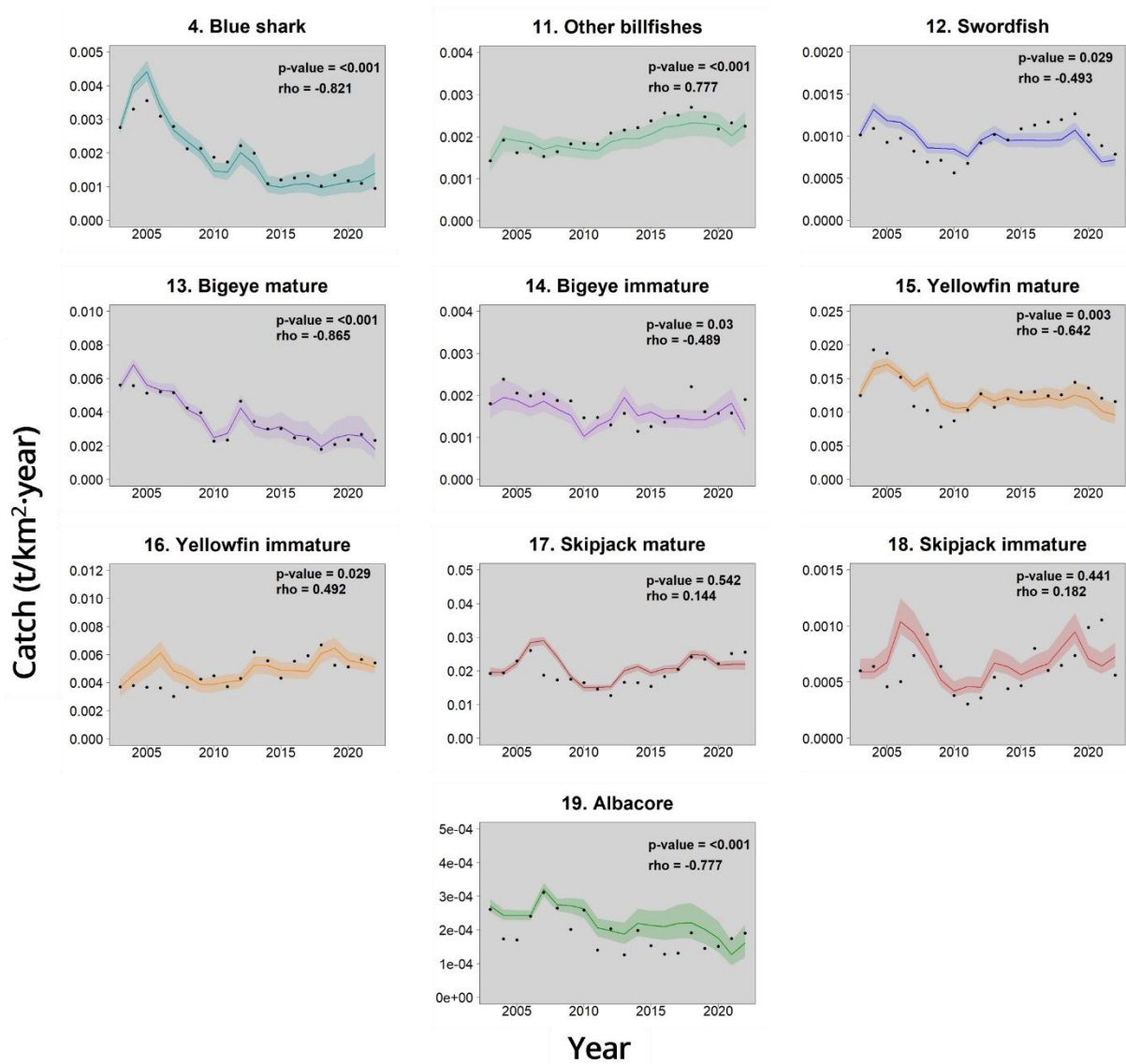


Figure 3. Predicted (solid line) versus observed (black dots) values of catch (t/km²·year) of the functional groups with catch time series for the period 2003 – 2022 in the TIO model. Coloured areas around the prediction line represent the 5th and 95th percentiles obtained by the Monte Carlo routine. The p-value and rho are the results of the Spearman correlation test.

3.2. Ecosystem indicators

The predatory biomass (Figure 4) exhibits a declining trend during the initial years of the simulation, and then gradually increases toward the end of the simulation period, but it does not recover to its initial value, remaining 8.7% lower (Table 6). The mean trophic level of the community (mTLco) and the mean trophic level of the community with a trophic level larger than 3.25 (Figure 4), remain stable although it experiences slight fluctuations throughout the simulation. Predictions for these three indicators display considerable uncertainty, and the Spearman correlation test yields a non-significant p-value. Conversely, the commercial species biomass (Figure 4) shows a negative trend throughout the simulated period, decreasing by more than 30% by the end of the period compared to its initial value (p-value <0.05) (Table 6). Similarly, both the mean trophic level of catches (mTLc) (Figure 4) and the mean trophic level of the community with a trophic level greater than 4 (mTLco>4) show a similar decline (p-value >0.05) (Figure 4). The mTLc decreases by 2.5% despite fluctuations, while the mTLco>4 show a more consistent decline of nearly 1% (Table 6). The mean trophic level of the community with a trophic level larger than 2 (Figure 4) also shows a small but steady decline of almost 1% at the end of the simulated period (Table 6). Finally,

the Kempton's Q index (Figure 4) also shows a stable, significantly declining trend, decreasing by more than 20% by the end of the simulation compared to its initial value (Table 6).

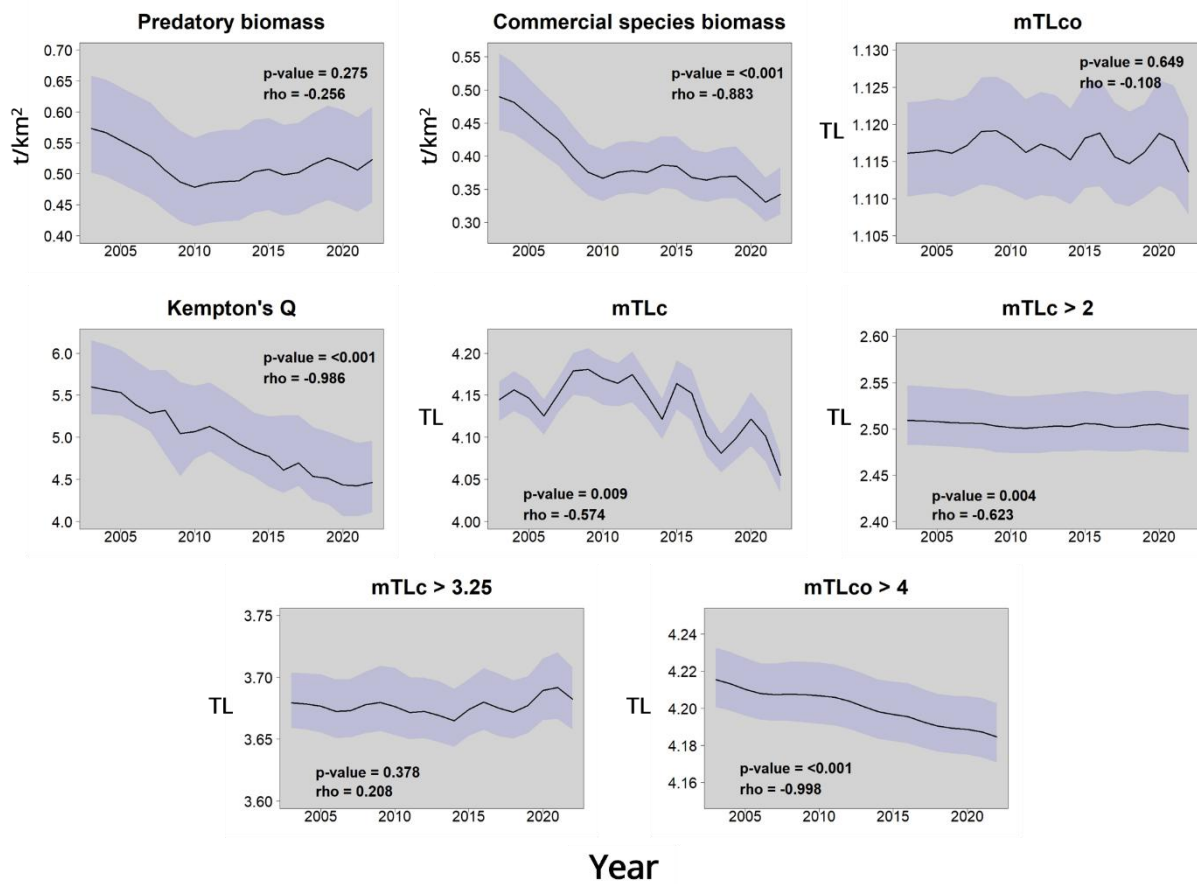


Figure 4. Predicted (black line) values of each indicator for the period 2003 – 2022 in the TIO model. Coloured area around the prediction line represents the 5th and 95th percentiles obtained by the Monte Carlo routine. The p-value and rho are the results of the Spearman correlation test. mTLco: mean trophic level of the community; mTLc: mean trophic level of the catch; mTLco > 2: mean trophic level of the community with a trophic level higher than 2 and lower than 3.25. mTLco > 3.25: mean trophic level of the community with a trophic level higher than 3.25 and lower than 4; mTLc > 4: mean trophic level of the community with a trophic level higher than 4.

Table 6. Values of the 8 selected indicators at the initial year (2003) and at the end of the simulated period (2022) and their change in percentage.

Indicator	Initial (2003)	Final (2022)	Change in %
Predatory biomass	0.57	0.52	-8.7
Commercial species biomass	0.49	0.34	-30.6
Mean trophic level of the catch (mTLc)	4.15	4.05	-2.5
Kempton's Q	5.60	4.46	-20.4
Mean trophic level of the community (mTLco)	1.12	1.11	-0.9
Mean trophic level of the community with a TL >2	2.51	2.49	-0.8
Mean trophic level of the community with a TL >3.25	3.68	3.68	0
Mean trophic level of the community with a TL >4	4.22	4.18	-0.95

4. Discussion

This study provides the first comprehensive ecosystem model of the tropical Indian Ocean oceanic pelagic ecosystem. Using the Ecopath with Ecosim approach, we integrated species interactions and the

ecosystem dynamics with fishing impacts using historical data from 2003 to 2022. The model adequately reproduced most observed time series of biomass and catches for most species and functional groups. For the three tropical tuna species, the model performs particularly well for mature stages but is less accurate for immature stages due to higher variability in the observed data. While other functional groups such as albacore, other billfishes, and blue shark are generally well modelled, the swordfish trend is less accurately captured. Catch predictions perform well for the ten functional groups with available fishing mortality time series but show poorer alignment with catch data for those without such time series.

Results reveal significant declines in biomass for several top predatory species, in line with some of the existing stock assessments. In contrast, species subject to lower fishing exploitation show more stable or increasing trends, highlighting differential responses of the functional groups across the food web. These findings suggest that human activities such as fishing along with environmental variability not only can affect target species but also cascade through the ecosystem altering, trophic structure, ecosystem functioning, and biodiversity.

By providing a holistic understanding of ecosystem dynamics and the cumulative effect of fishing on the ecosystem within the last 20 years, this work offers critical information to form ecosystem-based advice to support the management of tropical tuna species and associated ecosystems in the tropical Indian Ocean and guide EAFM implementation.

Fishing has shaped the biomass trends of tuna and tuna-like species and their associated ecosystem differently across the oceanic ecosystem of the tropical Indian Ocean. Species that have been overfished in recent years (e.g., other billfishes) generally continue to decline in biomass. In contrast, species that haven't been overfished (maintained above their B_{MSY}) (e.g., swordfish), or that have experienced a reduction in fishing mortalities (e.g., blue shark, skipjack tuna) in recent years, show increasing biomass trends in the final years of the simulation. This pattern is attributed to moderate fishing pressure and greater resource availability resulting from the decline in competitor abundance. These results seem to be in line with those from the assessments probably due to the use of their outputs to fit the model. Vulnerable species incidentally caught by various fleets (e.g., silky shark, other sharks, and mobulas) seem to exhibit a marked decrease in biomass due to the high mortality rates associated with bycatch events. Overall, these results suggest that the ecosystem may be negatively impacted by fishing, particularly at higher trophic levels and with clear effects on certain species. However, with appropriate resource management, population stability and recovery could be achievable.

Ecological indicators reflect further the impacts of fishing and trophic changes in the ecosystem. The decline in the biomass of commercial species at higher trophic levels, as well as the decrease in the mean trophic level of the catch may suggest an increasing proportion of immatures being caught relative to matures, or a shift towards harvesting species from lower trophic levels. The slight decrease in the mean trophic level of low-trophic-level species suggests a top-down trophic control—the decline of predators allows for an increase in mid-trophic-level species, which in turn leads to a reduction in the trophic levels immediately below. Furthermore, the pronounced decline in the Kempton's Q index, suggest a sustained reduction in species diversity within the ecosystem. These results coincide with those obtained for the historical period in the Ecosim model of the Western Pacific Warm Pool Province (Griffiths et al., 2018) revealing a reduction in species diversity at the higher trophic levels. Although in both cases the catches of target species decrease, in the Pacific they are replaced by species of similar trophic level, whereas in the Indian Ocean they are substituted by species of lower trophic level. However, in both scenarios, the replacement species are of no commercial interest to the fishery at present.

Ecosystem models are limited by availability and quality of data, which is the principal factor introducing uncertainty in model outcomes. Key challenges on information gaps involved collecting biomass estimates, catch and effort data and diet composition, which are basic input data for the model. This highlights the lack of biological and fisheries data in oceanic regions, particularly within the study area, and therefore, the need to increase their collection.

Biomass estimates are scarce across all trophic levels within the study area, particularly for species not targeted by major fisheries. This includes apex predators as well as pelagic invertebrates, and it limits the model's ability to fit time series data. Therefore, future efforts should focus on estimating and monitoring

the biomass of key species or functional groups in the ecosystem, using methods such as acoustic detection, pelagic fishing surveys, plankton sampling, or onboard observer programs. Dietary composition data were also insufficient often based on limited samples or highly localized studies in specific areas and time periods, making the data unrepresentative of the species as a whole or the intended temporal scope. This lack of high-quality biomass, catch and diet information becomes more pronounced for species at intermediate and lower trophic levels being generally poorly studied. To accurately characterize ecosystem structure and interactions, studies on the trophic ecology of species are essential. Given that stomach content analysis is complex and time-consuming, alternative methods such as DNA-based analysis of stomach contents may also offer a promising approach to deciphering the trophic ecology of species (Gunther et al., 2021, Canals et al., 2024).

Model calibration, validation, and uncertainty analysis are essential to ensure reliable model performance management, quality and relevance. The model developed in this study incorporates the best available information and follows best practices for ecosystem model construction and calibration (Heymans et al., 2014). For the development of the Ecopath TIO model, the PREBAL analysis (Link, 2010) was applied to ensure consistency of the basic input parameters with general ecological principles and to guide the balancing process. Nevertheless, the model and its projections present certain limitations as previously discussed. Although the model includes time series of biomass, catches, fishing mortality for key predators, and fleets' fishing effort, some of these time series have been derived based on significant assumptions. For example, blue shark, swordfish, and albacore tuna are present within the modelled area but their distribution ranges are broader and include the temperate zone of the Indian Ocean. Their dynamics within the model may therefore represent the tropical component of their populations but may not reflect the entire stock dynamics. A further limitation concerns fishing effort data for multiple fleets. Many fleets report effort inconsistently, only for certain years, in partial coverage or in different effort units and some do not report effort at all. These inconsistencies make it difficult to realistically estimate trends and approximate the variation in fishing effort across fleets. In addition, the model does not include a parameter to account for the unquantified variability in fishing efficiency over time due to the adoption of various adjustments in fishing practices—commonly referred to as 'effort creep'. Failing to account for this technological improvement may result in higher catch rates not being properly quantified, thereby compromising the reliability of the effort time series and the model's outputs.

The model also faces limitations in estimating bycatch and its subsequent fate. Although, the IOTC nominal catch database is intended to include both landings and discards across all fleets, in practice, most fleets report only their landings. Consequently, bycatch rates had to be inferred from published literature, and for a few fleets, estimated using onboard observers' data. This approach presents several limitations. The main limitation is that, in the absence of bycatch data, or in cases where there are significant differences in data quality, the model fails to reflect the actual impacts of certain fleets on vulnerable species and the ecosystem. This leads to a bias in the representation of bycatch effects, disproportionately attributing greater impact to fleets that report bycatch more comprehensively. This issue is particularly relevant in the case of purse seine fisheries using FADs, which have received considerable attention due to concerns about their ecological impacts, have been extensively studied, and for which bycatch data is largely available. However, it is essential to contextualize these concerns by comparing bycatch rates across different fishing gears. Empirical data indicate that FAD-associated purse seine fisheries often exhibit lower bycatch rates than other methods such as longline or driftnets (Garcia & Herrera., 2018; Murua et al., 2021). Therefore, a comprehensive evaluation of fishing fleets impacts should incorporate bycatch estimates within a reasonable level of uncertainty alongside catch efficiency for target species. Another limitation of the approach is that the estimates are expressed as a proportion of the target species catches for a specific fleet, year, or period. Applying these rates across the entire simulated period requires the assumption that bycatch rates remain constant over time. Additionally, there are limitations in estimating the discards' fate. To address this, we calculated the proportion of discards that results in mortality based on studies of at-haulback and post-release mortality. These studies are scarce and typically cover specific fleets and time periods, with limited sample sizes. Therefore, we assume that the proportion of bycatch mortality remains constant throughout the simulation, except for the Spanish, French, and Seychellois fleets. This assumption holds despite the fact that the applied value is representative of a specific period only, and notwithstanding ongoing efforts to enhance bycatch

survival through improved release practices. Assuming two distinct periods for discard survival rates—one prior to the implementation of best practices and another following it—with no transitional phase introduces substantial variability in catch estimates for the Spanish, French, and Seychellois purse seine fleets. This variability can reach orders of magnitude for functional groups whose catch time series rely primarily or exclusively on discards (e.g., baleen whales, whale sharks, pelagic stingrays, mobulas). Consequently, for functional groups lacking a fishing mortality time series, particularly those for which the adoption of best practices has significantly improved survival rates (Grande et al., 2019, Grande et al., 2022, Stewart et al., 2024), the model encounters difficulties in fitting the catch time series due to the abrupt change in the data. Nevertheless, for some functional groups such as the silky shark, the model was able to fit their catch time series despite these concerns.

Updating and refining the model will improve its value for management and future ecosystem assessments. As new information becomes available, the current EwE models can be updated and further developed, thereby enhancing their quality and increasing their value as tools for understanding impacts and exploring management options for the tropical Indian Ocean pelagic ecosystem. In the SPC and IATTC EwE models are being used to explore alternative fishing strategies and to assess the combined impacts of fishing and climate change on key species (Griffiths et al., 2021, Allain et al., 2007). These models provide valuable ecosystem-level insights that complement traditional approaches and inform discussions on sustainable fisheries governance.

5. Conclusions and future steps

Being able to model the tropical Indian Ocean ecosystem and its constituent organisms under the influence of fishing over the past two decades represents a significant step towards providing ecosystem-based advice for supporting EAFM implementation in IOTC. This study has developed an Ecopath with Ecosim model of the tropical Indian Ocean oceanic pelagic ecosystem to improve understanding of the ecosystem's dynamics, species interactions and the impact of fishing on the ecosystem structure and functioning. Results reveal variations in the biomass of higher trophic level species, with the greatest variations observed in those of highest commercial interest. The findings corroborate the results of the assessments and the fact that implementing the scientific advice may lead to improvements in biomass. Moreover, the study also emphasizes the ecosystem's complexity and the intricate trophic interactions within it, reinforcing the need to incorporate these dynamics into marine resource management in IOTC.

Further development of the model will allow more realistic fits and projections under a wide range of management scenarios. Future work could focus on incorporating technological advancements for each fleet or fishing gear type to more accurately simulate fleet effort trends. Additionally, to better understand the impacts of fishing fleets on vulnerable bycatch species, and in collaboration with the IOTC Secretariat, efforts will be made to identify the species considered vulnerable in the Indian Ocean. This will enable the development of indicators to assess the effects of fishing on vulnerable species and to certain taxa groups. Moreover, work will be carried out to incorporate the effects of climate change allowing for simulations of future scenarios under varying conditions of climate change, fishing pressure, and their combined effects, and to analyse the responses at ecosystem and species-level. Finally, to expand the range of scenarios and impacts that can be analysed, a time series will be developed to reflect the reduction in bycatch of certain species resulting from the reduction in FAD numbers and improvements in their design. These improvements will enhance the model's capacity to produce more realistic outcomes and to simulate possible future management scenarios. This will make it a more relevant and reliable tool for informing fisheries management, incorporating ecosystem considerations to guide EAFM implementation within IOTC.

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Supplementary material

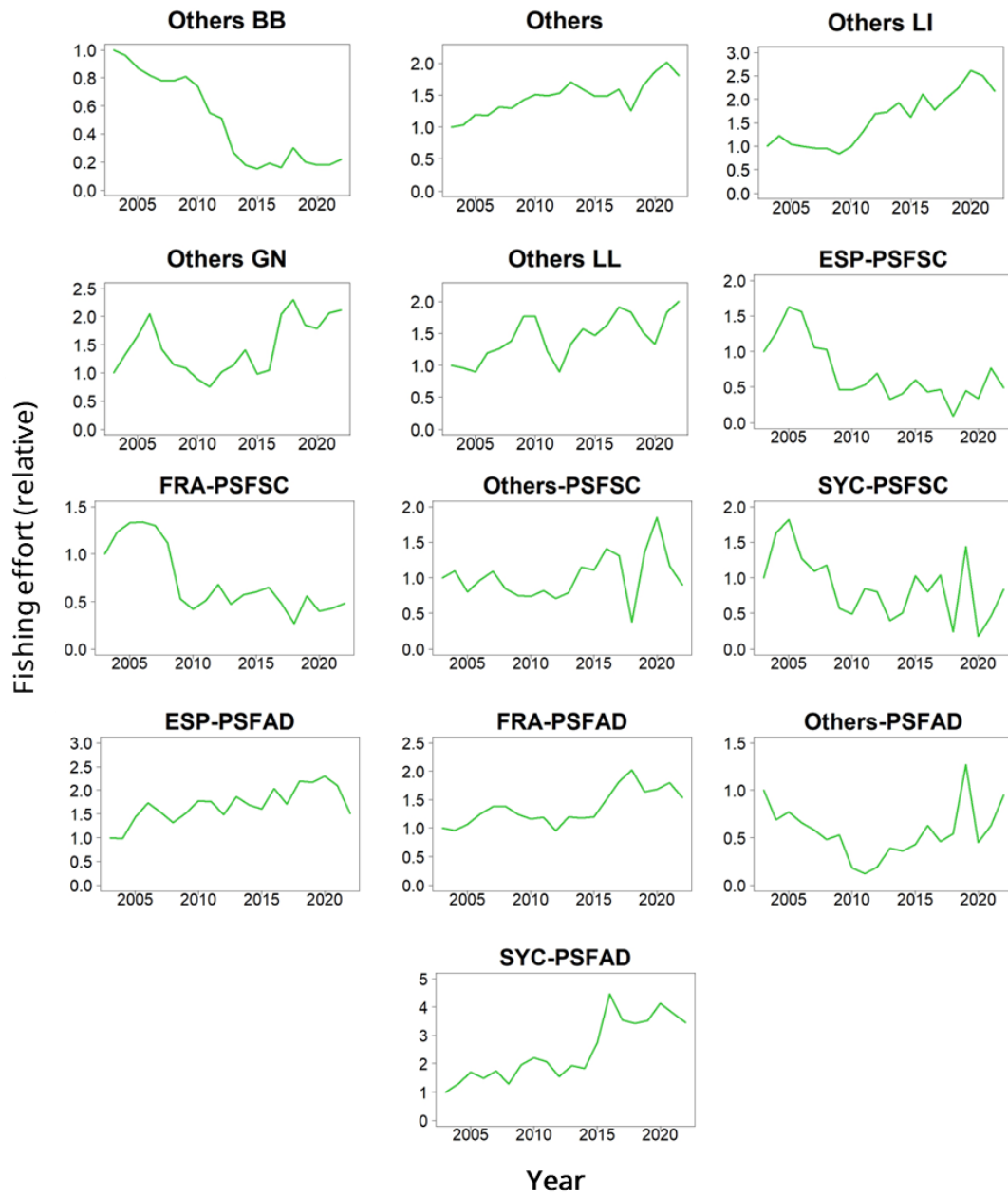


Figure 1. Time series of relative fishing effort of each fleet inside the model area along the historical period (2003-2022) used to drive the model.

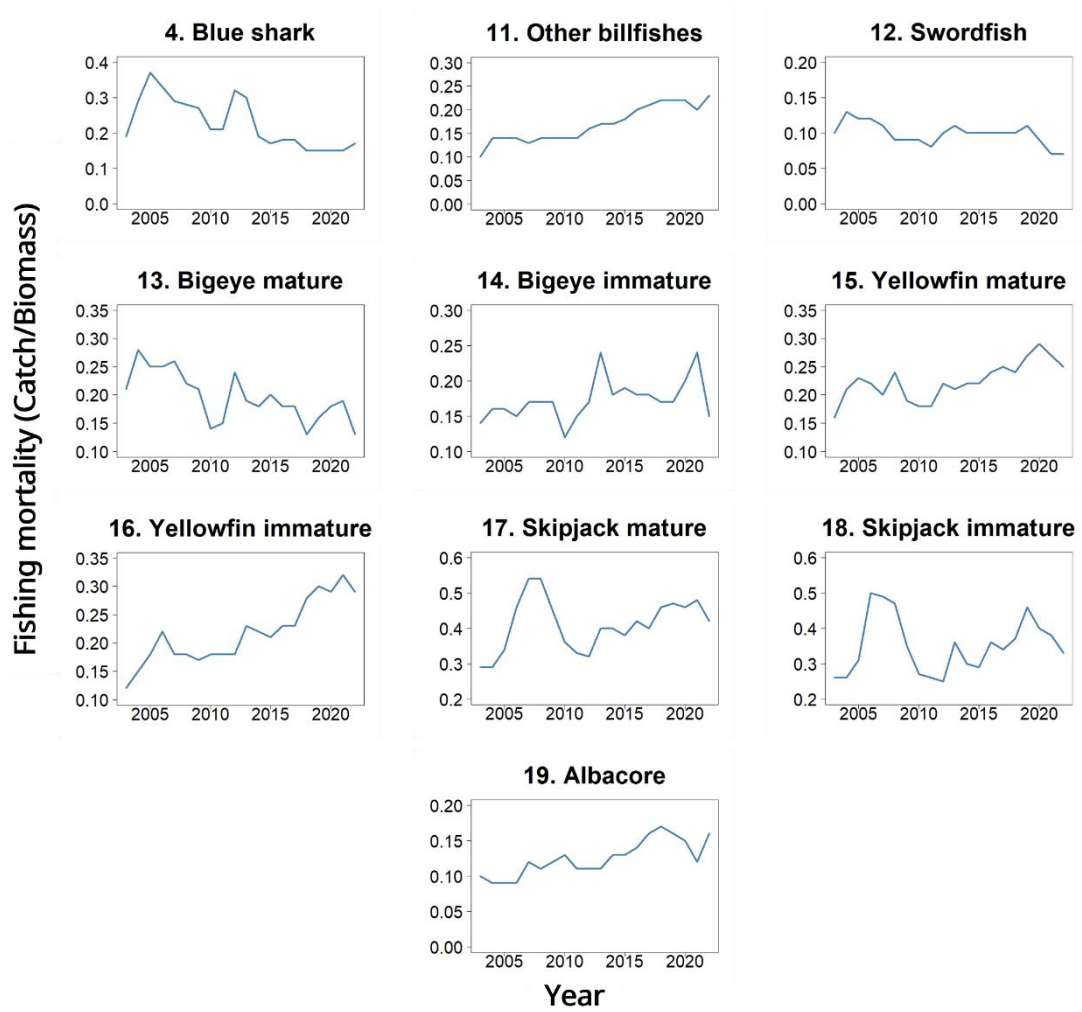


Figure 2. Time series of fishing mortality (catch/biomass) for the functional groups with available stock assessment.

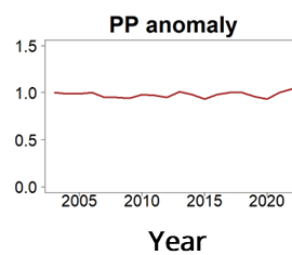


Figure 3. Time series of the relative variation of the primary production inside the model area along the historical period (2003-20022).

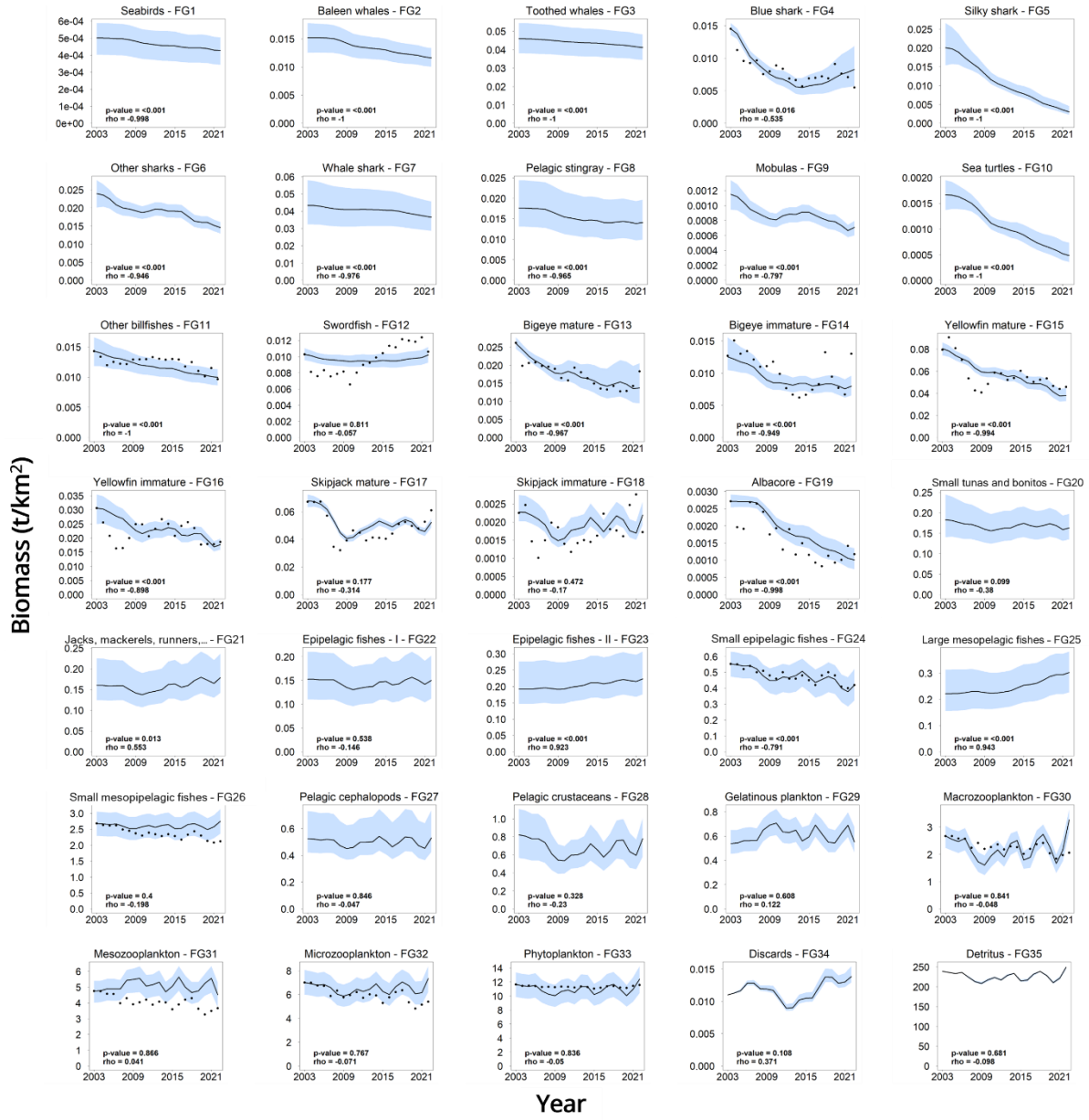


Figure 5. Predictions (solid lines) versus observations (black dots) of biomass (t/km²) for all functional groups in the TIO model during the historical period (2003–2022). The blue shaded areas around the prediction lines represent the 5th and 95th percentiles obtained using the Monte Carlo routine. The p-value and rho are the results of the Spearman correlation test.

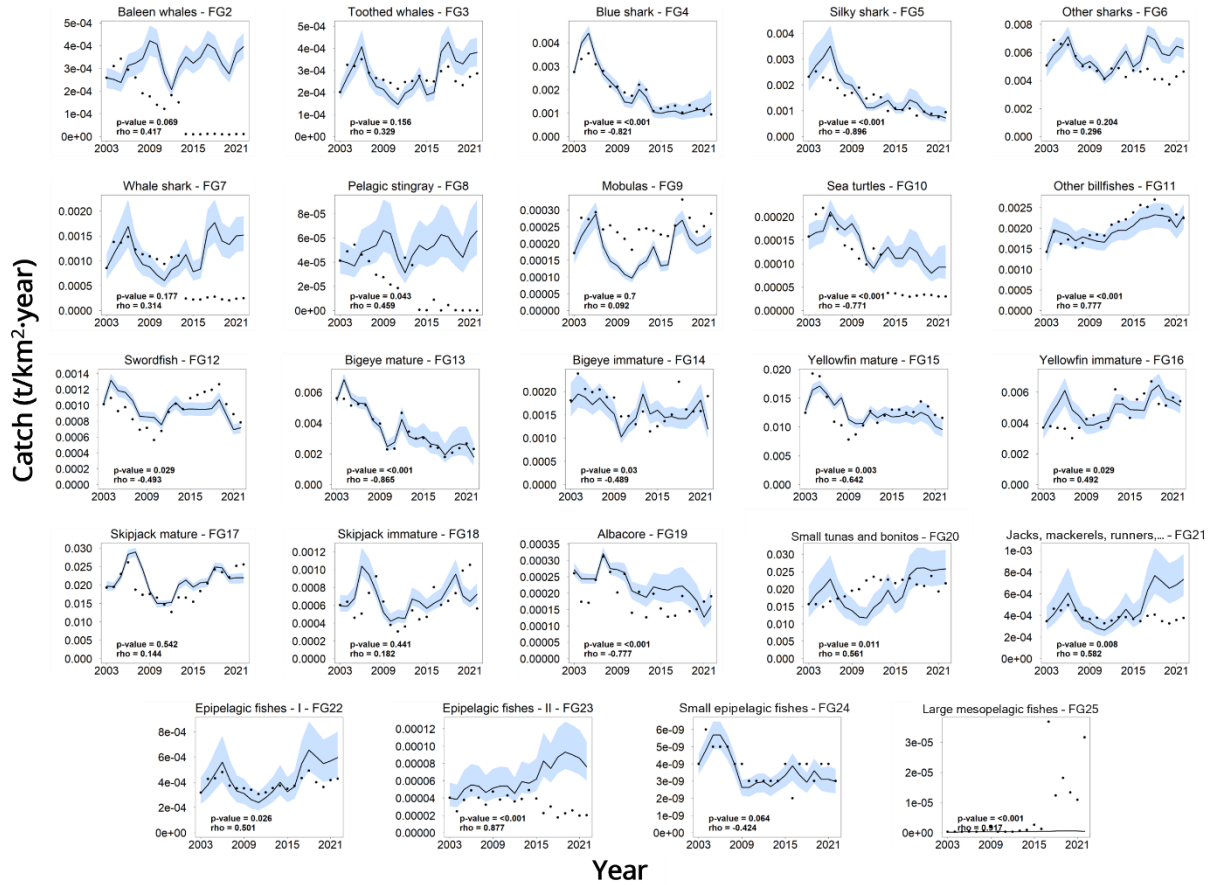


Figure 6. Predictions (solid lines) versus observations (black dots) of catch (t/km²-año) for all functional groups in the TIO model during the historical period (2003–2022). The blue shaded areas around the prediction lines represent the 5th and 95th percentiles obtained using the Monte Carlo routine. The p-value and rho are the results of the Spearman correlation test.